Mississippi Valley State University

# CS 341 — Discrete Structures

Practice Set 7: Approved Solutions

*Tautologies, Contradictions, and Rules of Inference*

Covers Chartrand & Zhang, Section 1.5

**Status of this set**

**Ungraded practice.** Solutions released with it. Attempt everything first. Fill truth tables on this page.

# Part I: How to Practice This Material

## Three outcomes, not two

**The classification**

Build the final column. Then:

- all T ⇒ **tautology**
- all F ⇒ **contradiction**
- a mix ⇒ **neither**

"Neither" is the usual case. Do not force a statement into one of the two extremes because the problem seems to want it.

**Three mistakes to expect**

1. **Stopping after two rows.** A tautology claim needs every row.
2. **Mistaking a fallacy for a rule.** Affirming the consequent looks like modus ponens and is not a tautology.
3. **Forgetting that an equivalence is a tautology.** If $R \equiv S$ then $R \Leftrightarrow S$ is true in every row.

# Part II: Practice Problems

## Section A — Classify

**1.** For each, build a truth table and classify it as a tautology, a contradiction, or neither.

(a) $(P \Rightarrow Q) \vee (Q \Rightarrow P)$

(b) $(P \vee Q) \Rightarrow (P \wedge Q)$

(c) $\big(P \wedge (\sim Q)\big) \wedge \big((\sim P) \vee Q\big)$

**Approved solution**

**(a) Tautology.** Column: T, T, T, T. Worth pausing on — it says that for *any* two statements, at least one of the two implications between them holds. That is startling until you remember an implication is false in only one row, and the two implications cannot be false in the same row.
**(b) Neither.** Column: T, F, F, T. True when $P$ and $Q$ agree, false otherwise — so this is just $P \Leftrightarrow Q$ wearing a different coat.
**(c) Contradiction.** Column: F, F, F, F. By Theorem 1.48 the right factor is $P \Rightarrow Q$, and the left says $P$ holds with $Q$ false, which is exactly when $P \Rightarrow Q$ fails. The two can never hold together.

**2.** Let $S$ be a tautology and $R$ a contradiction, built from the same components. Determine the truth value of each: (a) $S \vee R$; (b) $S \wedge R$; (c) $S \Rightarrow R$; (d) $R \Rightarrow S$; (e) $S \Leftrightarrow R$.

**Approved solution**

$S$ is true in every row and $R$ is false in every row, so every answer is the same in every row.
**(a) True** — a disjunction with a true part.
**(b) False** — a conjunction with a false part.
**(c) False** — true hypothesis, false conclusion, in every row.
**(d) True** — false hypothesis, in every row.
**(e) False** — the two never agree.
No table is required. Once you know both columns are constant, each answer follows from a single row of the relevant definition.

## Section B — Rules and fallacies

**3.** For each argument, say whether the reasoning is valid. If it is, name the rule; if not, name the fallacy and give a situation where the premises hold but the conclusion fails.

(a) If the server is down, the page fails to load. The server is down. So the page fails to load.

(b) If the server is down, the page fails to load. The page failed to load. So the server is down.

(c) If the server is down, the page fails to load. The page loaded. So the server is not down.

(d) If the server is down, the page fails to load. The server is not down. So the page loaded.

**Approved solution**

**(a) Valid — modus ponens.**
**(b) Fallacy — affirming the consequent** (assuming the converse). The page could fail for another reason entirely: a bad URL, a network outage, an expired certificate. Both premises hold and the conclusion fails.
**(c) Valid — modus tollens.**
**(d) Fallacy — denying the antecedent** (assuming the inverse). Same counterexample: the server is up but the page still fails for a different reason.

**(b) and (d) fail in the same situation**, and that is not an accident. They are the converse and the inverse, which Week 2 showed are equivalent to each other. One counterexample refutes both.

**4.** Verify that modus tollens, $\big((P \Rightarrow Q) \wedge (\sim Q)\big) \Rightarrow (\sim P)$, is a tautology.

| $P$ | $Q$ | $P \Rightarrow Q$ | $\sim Q$ | $(P \Rightarrow Q) \wedge (\sim Q)$ | Whole statement |
|---|---|---|---|---|---|
| T | T | T | F | F | T |
| T | F | F | T | F | T |
| F | T | T | F | F | T |
| F | F | T | T | T | T |

**Approved solution**

True in all four rows, so it is a tautology.
Notice how it works: the hypothesis $(P \Rightarrow Q) \wedge (\sim Q)$ is true in only one row — the last — and in that row $\sim P$ is also true. Every other row has a false hypothesis and is therefore true automatically. That is the general shape of these arguments. A rule of inference is a tautology because whenever the premises all hold, the conclusion holds too.

## Section C — Building to order

**5.** Find a compound statement built from $P$ and $Q$ that is true in exactly 50% of the rows. Then find one built from $P$, $Q$, and $R$ that is true in exactly 50% of its rows.

**Approved solution**

**Two components (2 of 4 rows).** $P$ itself works — it is true in rows 1 and 2. So does $P \oplus Q$, true in rows 2 and 3, and $P \Leftrightarrow Q$, true in rows 1 and 4.
**Three components (4 of 8 rows).** $P$ again works, as does $P \oplus Q$ or $P \oplus Q \oplus R$.

**The point of the problem** is that a compound statement need not use all its components, and the simplest answer is often a single letter. Students who spend ten minutes constructing something elaborate have solved a harder problem than the one asked.

## Section D — Writing

**6.** In five to eight complete sentences, explain why a rule of inference has to be a tautology, and why affirming the consequent is not one. Use the server example from Problem 3 concretely.

**Approved solution**

**Model answer.** A rule of inference licenses you to move from premises to a conclusion in *every* situation, so the implication from the premises to the conclusion must hold no matter what truth values the components take — that is exactly what it means to be a tautology. Modus ponens qualifies: $\big(P \wedge (P \Rightarrow Q)\big) \Rightarrow Q$ is true in all four rows, so whenever both premises hold, the conclusion does. Affirming the consequent, $\big(Q \wedge (P \Rightarrow Q)\big) \Rightarrow P$, fails in the row where $P$ is false and $Q$ is true.

Concretely: the server is up, and the page still failed to load because of a bad URL. Both premises of the argument are satisfied — the page failed, and it is true that a downed server would cause failure — yet the conclusion that the server is down is false. One such row is enough to disqualify it as a rule, because a rule must never lead from true premises to a false conclusion.

**Full credit requires** the connection between "valid in every case" and "true in every row." An answer that only reports the failing row earns about half.

## Section E — Challenge (optional)

**7.** Show that $\big((P \Rightarrow Q) \wedge (Q \Rightarrow R)\big) \Rightarrow (P \Rightarrow R)$ is a tautology. Then explain why this is *not* the same as claiming $(P \Rightarrow Q) \wedge (Q \Rightarrow R) \equiv P \Rightarrow R$.

**Approved solution**

The implication is true in all eight rows, so it is a tautology. It is the chaining rule (sometimes called hypothetical syllogism): if $P$ implies $Q$ and $Q$ implies $R$, then $P$ implies $R$.

**Why the equivalence is a different claim.** An equivalence requires the two sides to agree in every row. They do not: at $P$ true, $Q$ false, $R$ true, the left side is false while $P \Rightarrow R$ is true. They also disagree at $P$ false, $Q$ true, $R$ false.

So the chaining rule lets you conclude the right side *from* the left; it does not make them interchangeable. As you found in Practice Set 6, the textbook's Exercise 24(e) states the equivalence version, and that version is false.

**The general lesson:** $A \Rightarrow B$ being a tautology is a weaker claim than $A \equiv B$. Every equivalence gives two tautologies, but a single tautology gives only one direction.

**Instructor notes**

- **Problem 3 is the centre of the set.** Four arguments, same premise, two valid and two fallacious. Put the four on the board together — seeing them side by side is what makes the pattern stick.
- **Problem 3(b) and 3(d) share a counterexample.** Say so explicitly; it connects back to converse/inverse equivalence from Week 2 and makes the fallacies one idea rather than two.
- **Problem 5** rewards restraint. The answer is $P$. Watch for students who build something elaborate and never notice.
- **Problem 7** closes the Exercise 24(e) loop from Practice Set 6 by showing what the correct statement is. Worth pairing if you assigned that one.
- Reserved for Homework 4: Section 1.5 Exercises 2, 4, 6, 8, 10, 12 (even). Odd ones are self-check — answers are printed at pp. 596–597.